

TASTY, Kazakhstan

WMO: 380810

Lat: 44.800N

Lon: 69.117E

Elev: 623

StdP: 14.37

Time Zone: 6.00 (E06)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.3	-4.8	-15.9	2.4	-7.8	-11.7	3.0	-3.9	31.4	19.0	23.4	20.1	6.2	90	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.2	101.0	65.0	97.9	63.9	95.2	63.1	67.3	91.9	65.9	91.4	64.6	90.2	7.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.9	78.9	72.0	56.9	70.7	73.3	54.0	63.6	73.8	32.1	92.2	31.0	91.9	30.0	90.4	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.8	23.0	21.1	-13.9	107.7	7.7	4.1	-19.4	110.6	-23.9	113.0	-28.2	115.3	-33.8	118.3		
			-14.3	71.0	7.6	2.8	-19.7	73.0	-24.2	74.7	-28.4	76.2	-33.9	78.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	18.3	19.4	33.1	54.1	66.3	77.0	81.4	76.5	64.2	48.6	33.5	22.1	
	DBStd	24.49	11.13	11.49	11.06	8.93	8.10	6.71	5.46	5.53	7.50	9.24	10.34	12.20	
	HDD50	3941	984	858	532	59	6	0	0	0	2	136	498	866	
	HDD65	7032	1449	1278	990	339	78	4	0	2	104	513	944	1331	
	CDD50	3830	0	0	6	183	512	809	975	821	428	92	4	0	
	CDD65	1446	0	0	0	13	119	363	510	357	79	5	0	0	
	CDH74	20061	0	0	3	188	1428	5175	7385	4644	1145	93	0	0	
	CDH80	10608	0	0	0	43	535	2850	4345	2389	428	18	0	0	
Wind	WSAvg	7.6	7.3	8.7	9.1	9.6	8.1	7.0	6.4	6.9	7.3	7.1	7.4	6.8	
Precipitation	PrecAvg	6.2	0.7	0.6	0.6	0.7	0.8	0.3	0.5	0.2	0.1	0.4	0.8	0.6	
	PrecMax	11.2	1.1	1.8	1.5	1.8	2.4	1.5	2.2	1.1	0.4	1.4	3.0	1.8	
	PrecMin	3.6	0.2	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	
	PrecStd	2.2	0.2	0.4	0.4	0.5	0.7	0.4	0.6	0.2	0.1	0.4	0.9	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.3	48.2	68.1	85.7	95.0	102.6	106.6	102.4	93.8	82.1	63.2	49.1	
		MCWB	40.9	40.1	49.5	60.1	61.5	65.1	66.3	65.5	62.0	58.0	50.7	42.2	
	2%	DB	39.3	40.8	59.6	79.5	89.7	99.3	102.2	98.3	89.1	75.9	57.0	42.0	
		MCWB	35.7	36.1	47.2	56.7	60.6	63.9	65.6	63.7	59.5	55.1	48.3	38.1	
	5%	DB	35.4	36.3	53.3	74.7	86.0	96.6	98.9	94.9	85.0	70.2	51.7	38.5	
		MCWB	33.1	32.8	44.6	53.7	58.9	63.3	64.7	62.5	58.1	52.1	44.3	35.6	
	10%	DB	32.6	33.4	47.5	70.2	82.0	93.7	95.9	91.4	80.9	64.8	47.1	35.8	
		MCWB	31.0	30.9	40.9	52.2	58.0	62.3	63.9	61.5	56.4	49.4	41.5	33.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.3	40.5	54.7	62.2	65.3	67.4	70.7	67.4	64.2	59.4	52.1	42.3
			MCDB	45.3	47.5	65.0	78.8	81.8	94.5	93.4	90.7	89.2	78.4	59.1	47.3
2%		WB	36.1	36.1	48.4	59.1	63.2	65.6	68.2	65.0	60.5	55.8	48.8	38.4	
		MCDB	38.6	39.6	56.2	73.3	79.8	93.3	92.2	92.1	85.2	72.9	56.2	41.5	
5%		WB	33.2	33.1	44.7	56.1	61.5	64.4	66.8	63.6	58.6	52.9	45.4	36.1	
		MCDB	34.9	35.8	53.5	71.4	79.4	91.4	91.3	89.5	83.3	68.2	50.6	38.2	
10%		WB	31.0	31.0	41.1	53.5	60.0	63.3	65.6	62.5	56.7	50.1	42.2	33.7	
		MCDB	32.6	33.5	47.8	67.2	77.8	89.7	90.8	87.6	79.4	63.4	46.6	35.7	
Mean Daily Temperature Range	5% DB	MDBR	12.7	14.8	15.8	22.2	24.0	26.8	26.2	26.6	27.0	23.4	15.3	12.1	
		MCDBR	13.3	15.5	24.4	28.7	29.3	30.0	29.7	31.3	32.4	30.4	22.4	13.1	
		MCWBR	10.4	12.0	14.0	12.6	10.8	10.6	10.7	11.6	14.6	15.3	14.7	10.4	
	5% WB	MCDBR	12.4	14.5	22.2	24.4	24.0	27.1	26.3	27.1	30.6	28.5	19.4	12.7	
		MCWBR	9.9	11.6	13.5	13.1	9.8	10.2	9.8	10.8	14.4	15.0	13.6	10.3	
Clear-Sky Solar Irradiance	taub	0.352	0.384	0.434	0.470	0.471	0.450	0.439	0.406	0.376	0.390	0.382	0.352		
	taud	2.019	1.989	1.982	1.945	1.988	2.090	2.141	2.216	2.253	2.131	2.095	2.065		
	Ebn at Noon	225	242	248	251	255	261	263	267	266	239	213	210		
	Edh at Noon	34	43	49	56	55	50	47	42	37	36	32	30		
All-Sky Solar Radiation	RadAvg	475	799	1240	1735	2211	2410	2338	2120	1658	1048	587	391		
	RadStd	80	92	97	141	118	101	112	72	64	78	61	52		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page